

Muhammad Asif

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INTRODUCTION

Lieutenant in the Pakistan Navy with hands-on experience on submarines, working with diesel engines and generators, electric motors, air conditioning and ventilation, reverse osmosis plants, tanks, exhaust systems, monitoring systems and firefighting systems. Also an AI/ML engineer and full-stack developer, experienced in fine-tuning and deploying large language models with RAG, LangChain and agentic workflows, and in building FastAPI, React and Next.js applications. Works hands-on with embedded hardware, having designed two custom ESP32-based flight controllers for micro drones (coreless and brushless), covering schematic and PCB design, sensor integration, motor and ESC design, power management, wireless control and firmware. Seeking a master's in mechatronics to bring machinery, electronics and code together and take ideas from concept to working prototype independently.

EDUCATION

Pakistan Navy Engineering College, NUST, Karachi

2017 – 2021

BS Management Information Systems | CGPA: 3.36/4.00

TECHNICAL SKILLS

Marine & Mechanical Systems: Submarine machinery: diesel engines & generators, electric motors, HVAC & ventilation, reverse osmosis plants, tanks, exhaust and firefighting systems

Monitoring & Control: Shipboard monitoring systems, sensor-based feedback control

Backend: Laravel, Nest.js & FastAPI

Frontend: Blade, React, jQuery & Ajax

Databases: MongoDB, SQL & Redis

ML frameworks: PyTorch & Hugging Face Transformers

AI/ML Tools: Wandb (Weights & Biases), RAG, LangChain & n8n

Languages: HTML, CSS, JS, PHP & Python

Tools & DevOps: GitHub, Docker & CI/CD

Mechatronics & Embedded: Hardware–software integration, embedded firmware (ESP32), brushed motor drivers and brushless ESC design (AM32), battery power systems, LoRa wireless links

PCB & Electronics: Schematic capture and PCB layout in KiCad, DRC/ERC, fabrication outputs (Gerber, BOM, pick-and-place), component sourcing (LCSC/JLCPCB), prototyping & debugging

EXPERIENCE

Pakistan Navy

July 2021 – Present

Lieutenant (Submarine Service)

- Worked on submarine machinery and support systems, including diesel engines, diesel generators (DGs), electric motors and exhaust systems
- Worked with air conditioning, ventilation, tank and reverse osmosis (RO) water plant systems critical to onboard habitability
- Monitored and operated shipboard monitoring systems to track machinery condition and performance
- Developed in-depth knowledge of shipboard firefighting systems

Vital Analytics, US (Remote)

March 2025 – December 2025

AI/ML Engineer

- Fine-tuned and deployed multiple AI models, including Gemma, Qwen, Helsinki-NLP, Mistral and DeepSeek
- Built and deployed customised agentic AI solutions using RAG and LangChain
- Integrated the Model Context Protocol (MCP) into customised AI solutions
- Developed FastAPI backends connecting deployed models to React and Next.js frontends

MECHATRONICS PROJECTS

Micro Drones with Custom Flight Controllers (Coreless and Brushless) | github.com/Muhammad-Asif10710/KiCAD-Drone-Public

- Designed two ESP32-based flight controllers for 1S micro drones end to end in KiCad, from concept and schematic to PCB layout, fabrication files and firmware, without external support
- **Coreless version:** 2-layer board integrating sensing (MPU-6050 IMU), control (ESP32), actuation (four MOSFET motor drivers with flyback protection) and power (TP4056 onboard charging with automatic USB/battery power path and 3.3 V LDO)

- Built in USB-C programming with automatic bootloader entry and onboard charging, so the drone can be reflashed and recharged without extra hardware
- Implemented IMU-based stabilisation feedback so the controller reads orientation and adjusts motor output in real time
- Verified the board with DRC/ERC, trace-width and current checks, ground pours and stitching vias; produced Gerber, BOM and pick-and-place files for JLCPCB at about \$8 per board
- **Brushless version:** four onboard ESCs, each with an AT32F421 microcontroller running open-source AM32 firmware, an FD6288 three-phase gate driver and dual N-MOSFET half-bridges with sensorless back-EMF commutation; a boost converter powers the gate drivers from a single-cell battery
- Added a long-range 433 MHz LoRa control link (SX1278 Ra-02) over SPI, with parts selected for low cost and availability

SOFTWARE PROJECTS

- **Shonenshop** – Laravel- and React-based anime merchandise e-commerce platform
- **WebSocket Chat Application** – Real-time chat prototype with Stripe subscriptions and Twilio SMS/MMS integration
- **Housing Management Scheme** – Online platform for residential property, tenant, lease, rent and maintenance management
- **Confidential Public-Sector Projects** – Multiple web and AI systems (details under confidentiality)